DUNDALK INSTITUTE OF TECHNOLOGY
School of Informatics & Creative Arts
Ethical Approval Form for Research Projects

Researcher Name: **Vishnu Bhuvanachandran Nair Rajeswari Amma** Year: **2024-25**
Course: **M.Sc Data Analytics**

Title of project: **Investigations on Multi-Disease Prediction Framework for Dairy Cows Using Time-Series Behavioral Analytics**

Name of supervisor/s: **Dr. Zohaib Ijaz** Date: **29.03.2025**
(if applicable)

This application is to be completed by the researcher and where appropriate, in conjunction with the project supervisor. The lead researcher/supervisor is responsible for submitting the completed form to the appropriate Research Ethics Committee (details below)

Please note: If your submission is incomplete or unclear, your application will be returned to you and your project may be delayed.

Section 1

Type of Researcher

Please tick the appropriate box below to indicate the type of researcher you are:

Undergraduate ☐
(proceed to section 2)

Completed Ethical Approval forms for undergraduate research should be submitted to the relevant Departmental Research Ethics Committee (DREC).

Drec.dcamm@dkit.ie – Department of Creative Arts Media and Music
Drec.dcsm@dkit.ie – Department of Computing Science and Mathematics
Drec.dvhcc@dkit.ie – Department of Visual and Human Centred Computing

Postgraduate ☑
(proceed to section 3)

Completed Ethical Approval forms for Postgraduate research should be submitted directly to the School Research Ethics Committee (SREC).
Ica.srec@dkit.ie

Staff
(proceed to section 3) ☐

Completed Ethical Approval forms for Staff research should be submitted directly to the School Research Ethics Committee (SREC).

Ica.srec@dkit.ie

Section 2

Please complete questions 1-4 listed below.

	Human and / or Animal Research	YES	NO
1	**Does your research involve human participants other than the following[1]?** • Research using exclusively secondary sources. • Research using materials legally accessible to the public that have legal protection, e.g., record of court judgements, data archives. • Research using materials that are publicly accessible and where there is no reasonable expectation for privacy, e.g., books, published third party interviews. • Observations of human behaviour in public where (i) those being observed have no reasonable expectation of privacy, (ii) there is no intervention on the part of the researcher nor any interaction between the researcher and those observed, and (iii) individuals are not identifiable in the results. If '**YES**', please complete **B and C** below.		
2	**Does your project involve working with animals?** – If '**YES**', please complete **B and C** below. – Please note that for ethical consideration: 'Animals' are classed as vertebrate animals including cyclostomes and cephalopods (DIRECTIVE 2010/63/EU)		
3	**Does your project involve working with participants from any of the following categories?** • Minors (under 18 years of age) • People with learning or communication difficulties • Patients • People in custody • People engaged in illegal activities If '**YES**', please complete **D** below.		
4	**Does your project have any possible ethical implications other than those outlined in questions 1, 2 and 3?** If '**YES**', please complete **E** below.		

A. **I consider that this project has no significant ethical implications[2] to be brought through the ICA School Ethics Review Process**
Please complete Section 4

☐

B.

[1] If there is any doubt, researchers should contact the Chair of the SREC.
[2] In determining significant implications, please consider all potential risks attached to this project. If there is any doubt, researchers should contact the Chair of the SREC.

I. **Is this study part of a larger project that already has ethical clearance?**

YES / NO

If **YES** please answer question B II.
If **NO** please answer question C.

II. If this study is part of a larger project that already has ethical clearance, are you proposing any changes to the operational plan already ethically approved?

YES / NO

If **YES,** please complete *Sections 3 and 4.*
If **NO,** then please provide the project details below and complete *Section 4*.

Title of project with ethical clearance: _____

C. Could this project have ethical implications that should be brought before the appropriate ICA Departmental Ethics Review Committee as it will be carried out with human participants?

YES / NO

If **YES,** please complete *Sections 3 and 4.*
If **NO,** please complete *Section 4.*

D. I consider that this project may have ethical implications that should be brought before the School Research Ethics Committee as it will be carried out with human participants in a "vulnerable" category.
Please complete Sections 3 and 4 ☐

All research carried out with human participants in a vulnerable category must be referred by the Departmental Research Ethics Committee to the School Research Ethics Committee for approval.

E. Could this project have ethical implications, other than those previously outlined, that should be brought before the appropriate ICA Departmental Ethics Review Committee?

YES / NO

If **YES,** please complete *Sections 3 and 4.*
If **NO,** please complete *Section 4.*

Section 3

3.1 Application Form Checklist

Please complete Section 3 and provide additional information as attachments.

My application includes the following documentation:	**INCLUDED (mark as YES)**	**NOT APPLICABLE (mark as N/A)**
Recruitment advertisement		**N/A**
Participant Information Leaflet		**N/A**
Participant Informed Consent form		**N/A**
Questionnaire/Survey		**N/A**
Interview/Focus Group Questions		**N/A**
Debriefing material		**N/A**
Evidence of approval to gain access to off-site location		**N/A**
Ethical approval from external organizations. If ethical approval from external organizations is pending give details below		**N/A**
Details: **N/A**		

3.2 Project Details

a) Lay description (Maximum 200 words)

Please outline, in terms that any non-expert would understand, what your research project is about, including what participants will be required to do. Please explain any technical terms or discipline-specific phrases.

This research investigates the early prediction of diseases from cattle using a machine-learning approach. The analysis will be done using existing data sets from the literature. The disease prediction was carried out with respect to the diagnosis of lameness, mastitis, etc. The dataset was selected based on the studies reported by Lardy et al., 2024 from the French Governmental Agricultural Database, which was available to the public. The dataset contains the behavioral and associated disease data of about 300 dairy cattle concerning date and hour. With the help of time series analysis, we aim to predict multiple cattle diseases based on their behavioral characteristics.

b) Research objectives (Maximum 150 words)

Please summarise briefly the objectives of the research.

The study's main objective is to early predict cattle diseases using a machine learning approach using their daily routine data.

The specific objectives of the study are as follows:

- Selection of a database for the investigation from a suitable source
- Preprocess the data-data wrangling.
- Subject the data to feature selection and subsequent operations such as feature engineering.
- Perform the exploratory data analysis to analyze the trends, outliers, correlations, etc.
- Selection of suitable machine learning models for the most accurate prediction.
- Evaluation of performance.

Deploy the model for early prediction of cattle diseases with reasonable accuracy, which will benefit not only farmers but also veterinarians who aim to benefit the welfare of society.

c) Research location and duration

Location(s)/Population*	Dundalk Institute of Technology. DKIT
Research start date	01.03.2025
Research end date	01.08.2025
Approximate duration	6 Months

* If location/Population other than DkIT campus/population, provide details of the approval to gain access to that location/population as an appendix.

3.3 Participants

		YES	NO	N/A
Do participants fall into any of the following special groups?	**Minors (under 18 years of age)**			**N/A**
	People with learning or communication difficulties			**N/A**
	Patients			**N/A**
	People in custody			**N/A**
	People engaged in illegal activities (e.g. drug-taking)			**N/A**
Have you given due consideration to the need for satisfactory Garda clearance?				**N/A**

3.4 Sample Details

Approximate number	**N/A**

Where will participants be recruited from?	**N/A**
Inclusion Criteria	**N/A**
Exclusion Criteria	**N/A**
Will participants be remunerated, and if so in what form? **N/A**	

Justification for proposed sample size and for selecting a specific gender, age, or any other group if this is done in your research. N/A

3.5 Risk to Participants

a) Please describe any risks to participants that may arise due to the research. Such risks could include physical stress, emotional distress, perceived coercion e.g. lecturer interviewing own students. Detail the measures and considerations you have put in place to minimize these risks

b) What will you communicate to participants about any identified risks? Will any information be withheld from them about the research purpose or procedure? If so, please justify this decision.

3.6 Informed Consent

	YES	NO	N/A
Will you obtain active consent for participation?			**N/A**
Will you describe the main experimental procedures to participants in advance?			**N/A**
Will you inform the participants that their participation is voluntary and may be withdrawn at any point?			**N/A**
If the research is observational, will you ask for their consent to being observed?			**N/A**
With questionnaires, will you give participants the option of omitting questions they do not want to answer?			**N/A**
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?			**N/A**

Will the data be anonymous?			**N/A**
Will you debrief participants at the end of their participation?			**N/A**
Will your project involve deliberately misleading participants in any way, or will information be withheld? **If you answer yes, give details and justification for doing this below.**			**N/A**
N/A			

a) Please outline your approach to ensuring the confidentiality of data (that is, that the data will only be accessible to agreed upon parties and the safeguarding mechanisms you will put in place to achieve this.) You should include details on how and where the data will be stored, and who will have access to it.

The data will be collected from freely available online sources such as Journals, Websites, etc. This data will be collected, sorted, and stored in the DKIT One Drive and DKIT Gitlab. The project supervisor, the project committee, and I will have access to it. It should be noted that the datasets are publicly available, and the specifics related to my dissertation will be stored in DKIT OneDrive and Gitlab, to which the project committee will only have access.

b) Please outline how long the data will be retained for, if it will be destroyed and how it will be destroyed.

1. **Storage : DKIT Gitlab, DKIT One Drive, (7 Months)**
2. **Access : Supervisor and the Dissertation committee**
3. **Communication: Direct Reporting**
4. **Digital Platform Usage: DKIT Gitlab**
5. **Data maintenance: N/A**

Section 4

Researcher I have read and I understand the DkIT Ethics Policy available from: https://www.dkit.ie/assets/uploads/documents/Research/Policies/DkIT%20Research%20Ethics%20Policy.pdf

Signed: Print Name: **Vishnu Bhuvanachandran Nair Rajeswari Amma**

Date**: 29.03.2025** (Researcher)

Supervisor: Applications for Ethical Approval of Undergraduate projects are forwarded to the Departmental Research Ethics Committee for approval or referral to the School Research Ethics Committee. Applications for Ethical Approval of Postgraduate and Staff projects are sent to the School Research Ethics Committee for Approval.

I have read and approved this form & information:

Signed: _____ _____Print Name:**Dr. Zohaib Ijaz**

Date:**29.03.2025**

(Supervisor/Head of Department/ Research Centre Director/ Head of School)

There is an obligation on the researcher and/or supervisor to bring to the attention of the Departmental/School Research Ethics Committee(s): (a) Any issues with ethical implications not clearly covered by this form (b) Any ethical issues which may arise during the carrying out of the research; (c) Any ethically significant change made to the project after approval.

Section 5 (For office use only)

STATEMENT OF ETHICAL APPROVAL (FOR UNDERGRADUATE PROJECTS ONLY)

This project has been considered using agreed department procedures and is now:

Approved: ☐

Referred to the School Ethics Committee: ☐

Signed: _____Print Name: _____Date: _____
(Chair of Departmental Research Ethics Committee/Head of Department)

STATEMENT OF ETHICAL APPROVAL

This project has been considered using agreed School procedures and is now:

Approved: ☐

Rejected (further information sought): ☐

Chair of School Research Ethics Committee
This project has been considered by the Ethics Committee and ethical approval is granted.

Signed: _____Print Name: _____Date: _____
Chair of School Research Ethics Committee